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Charles E. Kahn, Jr., MD, MS
Editor, Radiology: Artificial Intelligence
Radiological Society of North America

Dear Dr Kahn,

Title. “Weakly Supervised Localization Generalizes Where Genotype Prediction Does Not: An Externally Validated Null for EGFR Radiogenomics from Lung CT.”

Authors. [Complete author list, in final order, with degrees.]

Article type. Original Research.

We submit this manuscript for your consideration. The CT radiogenomics literature reports EGFR prediction at AUC 0.80–0.89, almost always from single-center cohorts without external validation. We took a grounded vision–language model whose attention is supervised against tumor segmentations and separated two claims that this literature routinely bundles: that such a model localizes the lesion, and that it genotypes it. Localization transferred to 420 patients in another country, on another scanner fleet, in another treatment setting, beating a per-fold spatial shuffle, a center prior and an untrained head. Genotype prediction did not: pooled AUC was 0.618 against 0.794 for smoking status alone, adding the model to a five-variable clinical model changed AUC by -0.029 , no fusion topology helped, and calibration was worse than predicting the base rate.

We think the useful result is the dissociation rather than the null. A saliency map that tracks the tumor on an unseen institution’s data, attached to a classifier performing at 0.618, is evidence that plausible localization is not evidence of predictive validity — and that can only be observed when the prediction fails, so it cannot appear in studies that report both as successes.

Two objections we would rather raise than have noticed. First, the internal cohort is 153 patients with a known EGFR call and the pre-specified primary analysis set is 97, far below what the journal typically carries. Our argument is not that this is adequate but that the question is not resolvable by enlarging it: Riley’s criterion requires 414 patients for the five-variable clinical model alone, the entire open world of EGFR-labeled CT gives a 95% confidence-interval half-width of 0.154, and the largest external study yet published (1646 patients, 11 473 lesions, Eur Radiol 2026) reproduces our interval at ten times the sample size with our internal cohort as its external one. Second, the primary endpoint clears its pre-specified threshold by a single fold (11 of 15). Table 3 therefore reports every evaluation individually; the distribution is bimodal rather than marginal, with the weakest success at 0.757 pointing accuracy and the strongest failure at exactly 0.000.

Subject overlap with prior work (required). Both imaging collections are public and have been analyzed by other groups, including in each collection’s source publication and in the multi-tumor

external-validation study cited above, in which NSCLC-Radiogenomics serves as the external cohort. No patient in this analysis overlaps with any published, in-press, or under-review work by these authors. What is new here is the pre-specified separation of the localization and genotyping claims, the three-control external localization evaluation, and the measured precision bound. This overlap is also reported in Materials and Methods. [If any overlapping article exists, upload its PDF at submission and describe here what is new and different.]

Conflicts of interest and industry support (required). [REQUIRED — state all conflicts and any industry support, or state that there are none. ICMJE disclosure forms are submitted for every author.]

Sole submission (required). This manuscript is original, has not been published in the same or similar form by the authors, and has not been and will not be submitted to another journal by the authors or by colleagues at their institutions while it is under consideration at Radiology: Artificial Intelligence. [State here if any similar material has been provided to news or advertising media, and supply a copy.]

Use of artificial intelligence in the study and in manuscript preparation (required by RSNA Editorial Policies). [REQUIRED — complete and verify before submission.] Generative AI assistance was used in this work and is disclosed here and in the manuscript. Tool: [name, e.g. Claude Code with the Claude Opus model]; version: [exact model or release identifier]; manufacturer: [e.g. Anthropic, San Francisco, Calif]; dates of access: [range]. It was used for [state precisely which of: writing and refactoring analysis code; drafting and editing manuscript text; literature search; figure generation]. All analytical decisions, the pre-specified analysis plan, the interpretation of results, and the final text were reviewed and approved by the authors, who take full responsibility for the integrity and accuracy of the work. No AI tool is listed as an author.

Control of the data and the analyses (required by RSNA Editorial Policies). [REQUIRED — name the author(s) who had full access to and control of the data and who performed the statistical analyses, e.g. “[Initials] had full access to all study data and takes responsibility for the integrity of the data and the accuracy of the analysis.”]

Dual first authorship. [Delete if not applicable. Otherwise state which two authors contributed equally and why equal contribution is warranted; this must also appear on the full title page.]

Prior presentation and preprint posting. [REQUIRED disclosure at submission — state any presentation at a scientific meeting and any preprint posting with server, DOI and date, or state that there has been none.]

Reproducibility, for the editorial office. Every number in the manuscript is regenerated from archived out-of-fold predictions by a single script, requiring no imaging data and no GPU. The repository holds the pre-specified analysis plan with its deviation log — including the deviations that went against us — the completed CLAIM checklist, and all archived runs, negative results included. Because review is double-anonymized, the manuscript cites an anonymized mirror; the permanent URL is [REAL REPOSITORY URL] and the archival DOI is [ZENODO DOI], given here for the editorial office only.

Image licensing. NSCLC-Radiogenomics is licensed CC BY 3.0, and its images appear in Figure 6 with attribution. NSCLC-Radiomics is licensed CC BY-NC 3.0, so no pixel data from it appears in any figure — only derived statistics.

Ethics. [REQUIRED — insert the institutional review board’s written determination. The analysis

used only publicly available, fully de-identified data distributed by The Cancer Imaging Archive under open licenses, and no protected health information was accessed.]

Yours sincerely,

[Corresponding author name, degrees]
on behalf of all authors